ZERO CARBON

REFIRE

The whole is only
ever truly greater
than the sum of
its parts when
the result serves
the betterment of
mankind and
the planet

- Anonymous

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REFIRE STORY

PURPOSE

SO THE WORLD CAN BREATHE EASY >>

REFIRE Technology (REFIRE) was founded in 2015 and today the company is a leading international supplier of hydrogen fuel cell technologies. Named after the ideology of re-inventing fire, our core purpose is to transform the way people use energy across all walks of everyday life.

We believe in a future where zero-emission hydrogen-electric mobility and power will help to address the most urgent pollution and climate challenges, so the world can breathe easy.

To achieve this, we are accelerating mass adoption of hydrogen fuel cell products and services supported by sustainable ecosystems of value creation for both customers and users.

We aim to further strengthen our position at the forefront of the burgeoning global hydrogen-electric industry as a world class fuel cell technology innovator and builder.

We cannot achieve these goals alone. So our approach is to invite partners along for the ride, from policymakers to OEMs and end users, because the clean energy revolution will be successful when we work together.





CULTURE

DANCE LIKE A BUTTERFLY, STING LIKE A BEE >>

REFIRE is a technology and engineering company with the agility and character of a mature startup. Over the past seven years, we have attracted some of the brightest minds in new energy, automotive engineering, power electronics, stack science and manufacturing. Today, the REFIRE group of companies is over 650-strong including more than 280 engineers.

INNOVATION AT OUR CORE >>

Currently, China is home to the world's largest share of commercial fuel cell electric vehicles (FCEV's), and the 4,200+ fuel cell vehicles we've enabled to be deployed there equates to a domestic market share of almost 40%. The cumulative mileage of these vehicles already exceeds 145 million kilometres, representing a kerbside saving of around 83,000 tonnes of atmospheric carbon emissions[1].

What got us here, and what continues to drive us, is a blend of continual innovation and grit, together with a collaborative spirit that we share with all our customers and collaborators.

Our corporate culture emanates from the central vision of the company which is to reveal and catalyse clean and sustainable energy usage paradigms for the benefit of society and the earth.

[1] To Jan 31st 2023

STRATEGY

A SECURE AND INCLUSIVE ENERGY LANDSCAPE >>

REFIRE was founded with the conviction that hydrogen, as an energy carrier, would be viable for clean delivery, effective storage, and use case plurality.

With over 140 OEM-commissioned fuel cell powertrain prototypes under our belt - vehicles, drones, and powered machinery - that conviction aligns with real world requirements that will make hydrogen-electric power a feature of daily life - one in which energy is distributed more fairly, is less susceptible to supply shock, and is not wasted when demand falls.

Together with fuel cell technology, hydrogen is set to provide this more secure energy landscape.

ENABLER OF HYDROGEN ECOSYSTEMS >>

We have chosen to focus on optimisation of technological feasibility as well as commercial sustainability over the longer term. Early on, we recognised that a clean energy landscape would only arise from building end-to-end solutions that consider the full circle of energy and power from source to end use case.

As an early mover in the domestic fuel cell technology industry, REFIRE is positioned as a trusted intermediary that is developing the hydrogen value chain by clustering industry and resources, and by catalysing market opportunities.

Internationally, we are continuing to expand our activities in Europe and North America. Our customers there are benefitting from the real-world experience we've gained from the thousands of fuel cell-powered projects we've deployed that are operating daily in 20+ cities across the globe. We are also working with industry associations, think-tanks, and governments, to grow hydrogen economies that will support clean and sustainable mobility and power needs.

YOUR TRUSTED PARTNER >>

We set out to get things moving as quickly as possible, working with existing mobility and power infrastructure, and tackling the most polluting sectors first.

We aim to help our customers bring their fuel cell-powered projects to fruition on any timeframe. While vehicle retrofit customers may require turnkey solutions to meet very tight production schedules from as little as a few months, vehicle OEM's may benefit from more extensive system customisation, integration, and application engineering.

Our core components, including stacks, MEA (Membrane Electrode Assembly), and power electronics such as DCDC converters are also helping to satisfy customer needs at various stages of their own product development cycles.

Flexibility on top of reliability, durability, and above all, safety, is helping us to solve more customer challenges and meet more of their immediate needs.



REFIRE ASTRAWORKS MANUFACTURING INNOVATION CENTRE, CHANGSHU

VOLUME DELIVERY >>

Broad application at scale is essential. Anything less will fall short of the contribution required to effectively mitigate atmospheric pollution and climate disruption.

Quality as scale requires a level of consistency in production that has been a challenge for many fuel cell makers. To address this problem, we've concentrated our efforts since day one, on not only designing world-class products but solving the downstream manufacturing challenges ahead of time.

The result of this dual approach is our ASTRAWORKS manufacturing innovation and factory system. TüV certified, this semi-automated state-of-the-art facility will be iterated and duplicated where industry clustering and market forces demand it.

STANDARD STETTING >>

REFIRE is actively involved with global standards bodies to develop testing frameworks and benchmarks on a per-market basis to deliver location-specific solutions.

We are proud to have partnered with top global benchmarking, testing institutions to validate and certify system and stack performance:

- TüV Rheinland, Germany
- TüV Nord, Germany
- RDW, Netherlands
- GB (Guobiao) Standards, China

REFIRE is also fully certified for its quality management systems implementation through ISO and IATF standards:

- ISO9001:2015
- ISO14001:2015
- ISO45001:2018
- ISO26262:2018
- IATF16949:2016













THE HYDROGEN OPPORTUNITY

HYDROGEN: A QUANTUM SHIFT >>

A shift towards decarbonisation through policy, investment incentives, regulation, and consumer demand, is driving the new hydrogen economy. Hydrogen is now globally recognised as a clean, new energy carrier. More than 30 countries have released roadmaps for its widespread use.

Momentum is building along the entire value chain, bringing cost reductions to hydrogen production, distribution, and end user applications.

The net benefits of this are already catalysing technological innovation, projects, value, and ultimately clean air. We are hydrogen value chain enablers, engaging proactively with customers and collaborators, stakeholders, policymakers, industry advocates and, very importantly, the public.

LOCALISED ECOSYSTEMS >>

Let us be your facilitator of change, to help you benefit from the momentous opportunity to decarbonise and to re-invent the collective energy landscape.

We can support your growth through the strategic interlocking of technology, market forces, and policy.



READY TO GO >>

We've built over 140 OEM-commissioned, turn-key fuel cell powertrain prototypes enabling fast ramp-up to to deploy fuel cell mobility and power projects at scale.

From drones to passenger vehicles, trucks and buses to airport baggage handling trucks and mega-machinery such as port container movers, we have the right solutions for your application engineering challenge.

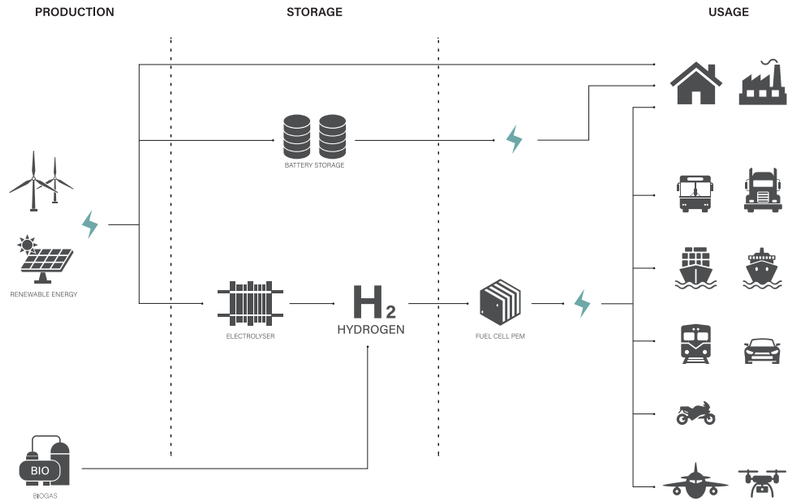
This unsurpassed catalogue of proven end use applications is the foundation of our expertise and the reason we are able to help our customers move more rapidly towards rolling out fleets of zero emissions vehicles into the real world.

ENTERPRISE SCALE KOL >>

With every step forward, the hydrogen economy is exploring uncharted territory. At this relatively early stage, the demand for authentic, balanced, and experienced industry voices to guide growth could not be more important.

REFIRE is sought not only for its class-leading products and services but also as a consultative voice across connected industries.

HYDROGEN ECOSYSTEM >>





OUR EDGE >>

PEOPLE
We are proud to say that REFIRE employs some of the sharpest minds in the industry. We are also actively pre-empting growth through headcount preparedness and skillset expansion as the market evolves.

ENGINEERING CULTURE
Engineering is in our blood. The company thrives on a healthy mix of innovation-driven and process-driven momentum. This gives us the confidence to tackle increasingly large scale and mission-critical projects.

BEST IN CLASS PRODUCTS & SERVICES
Our catalogue of world-firsts and standard-setting has established the company in an industry leadership role.

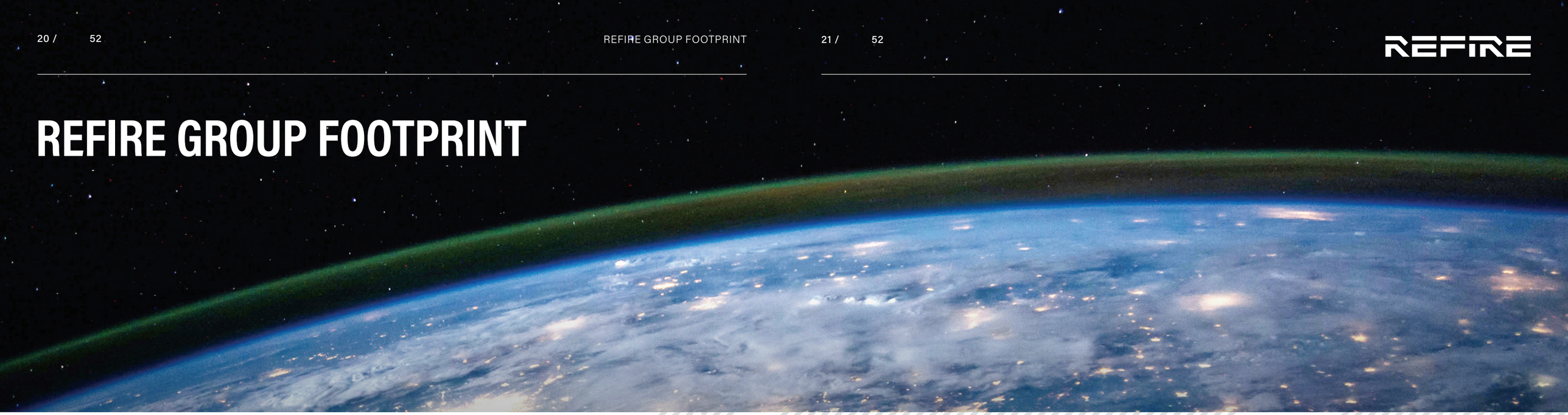
ECOSYSTEM OF PARTNERSHIPS
We continue to forge bonds with many of the world's finest global and local partners so that we can deliver on our promises.

DATA-DRIVEN APPROACH
Data is the fuel for our innovation and corporate evolution. It also allows us to offer customers a friction-free and highly optimised user experience.





REFIRE GROUP FOOTPRINT



ENTERPRISE SCALE >>

REFIRE is actively expanding its footprint domestically and internationally in regions where policy is driving industry strategy and where commercialisation can take place with fewer obstacles.

We would like to help your region develop optimised policy choices, drive strategy, and build market opportunities for the benefit of the planet.



1. BEIJING
FCS[1] Test & Certification

2. CHANGSHU
FCS Factory

3. SHANGHAI
R&D | Validation | Prototyping

4. FOSHAN
FCS Component Production

5. HONG KONG
International Sales

6. VANCOUVER
Stack R&D Centre

7. SWITERZERLAND
European Continent HQ[2]

8. SINGAPORE
SEA Sales Office[3]

[1] FCS (Fuel Cell System)
[2] Opening 2023
[3] Opening 2023

MOVE. BREATHE. EASY

4,200+ VEHICLES DEPLOYED[1]
4,500+ SYSTEMS DELIVERED
145,000,000+ KILOMETRES DRIVEN[2]
4,100+ TONNES OF H_2 CONSUMED[3]
140+ PROTOYTPE PROJECTS BUILT[4]
83,000+ TONNES CO_2 SAVED[5]

1 With REFIRE systems & products
2 To Jan 31st 2023
3 Estimate based on system power rating, to Jan 31st 2023
4 OEM-commissioned prototypes and production models built
5 Roadside carbon emissions, to Jan 31st 2023



PRODUCTS & SERVICES

PRISMA FUEL CELL SYSTEM SERIES

BEST IN CLASS >>

REFIRE produces the PRISMA line of fuel cell systems with benchmark models that are offered in multiple configurations. Each base system has extraordinary design flexibility to cater to very specific end use cases.

Launched in 2020, PRISMA is an industry-leading series of integrated fuel cell systems capable of powering everything from compact commercial vehicles up to long haul trucks.

USE CASE PLURALITY >>

PRISMA is ideally suited as a turnkey solution in the following sectors:

- MASS TRANSIT
- LOGISTICS
- UTILITY VEHICLES

Additional bespoke solutions based around PRISMA are available for:

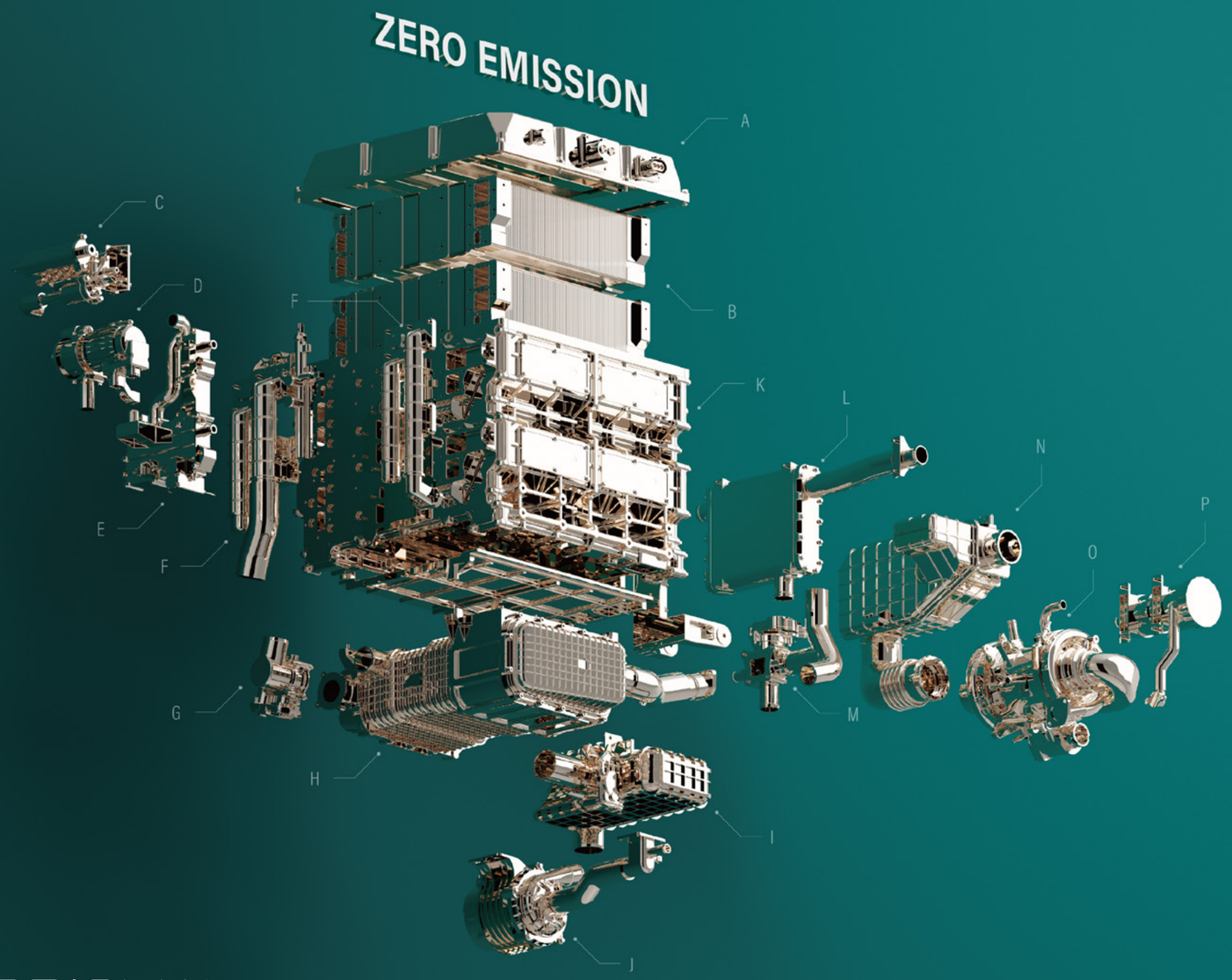
- AUTONOMOUS POWER
- DECENTRALISED GRID POWER

PRISMA BENCHMARK MODELS >>

PRISMA VI+
Light to medium-duty FCS that brings zero emissions to city logistics, mass transit, and urban services fleets.

PRISMA XII+
Heavy-duty FCS with improved torque, bringing higher top speeds and drive performance to freeway long-haul freight and intercity mass transit.





PRISMA®
Fuel Cell System Series

THE GOLD STANDARD FOR FUEL CELL ELECTRIC POWER

A DCDC with ACP & HRB controller
B Stack assembly
C Hydrogen jet + ejection assembly
D Hydrogen recirculation blower
E Water knockout
F Intake manifold assembly
G Throttle body
H HUM + intercooler
I Exhaust muffler
J Coolant pump
K Stack enclosure
L Heat exchanger
M Coolant thermostat
N Intake manifold
O Air compressor
P Deioniser

DURABILITY >>

REFIRE fuel cell systems are designed for rigorous commercial vehicle operating conditions. Depending on the exact model and user profile, our systems have durability design targets between 12,000 and 25,000 hours with voltage reduction of less than 10%.

FUEL ECONOMY >>

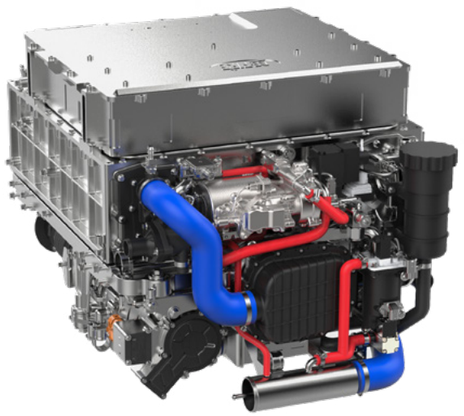
TCO (Total Cost of Ownership) is a key matrix for all commercial vehicle owners and operators.

REFIRE's fuel cell systems deliver one of the best fuel economies in the industry because we understand the importance of minimising operating costs for our customers through long term fuel savings.

SUSTAINABILITY >>

Once a fuel cell system can no longer be refurbished and reaches end of its usable life, it is easily dismantled, and a significant portion (by weight) can be recycled. Fuel cells are minimally toxic to take apart, and few special precautions are required for handling and disposal.

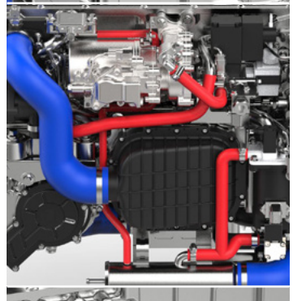


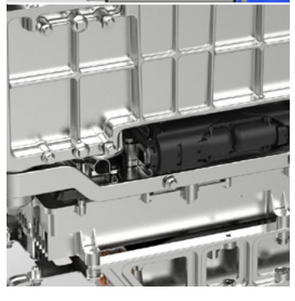


PRISMA VI+
Fuel Cell System

PRISMA VI+







DESCRIPTION

60kW Fuel Cell System
Light/Medium-Duty Applications

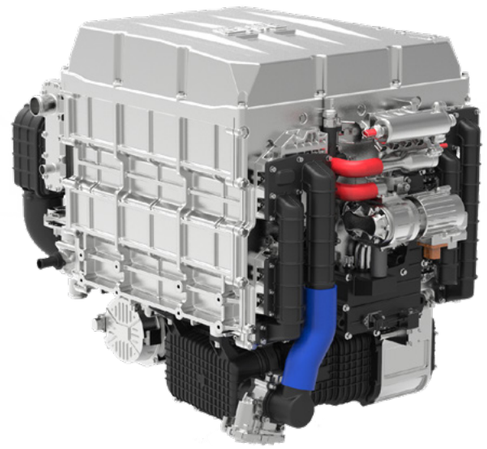
FEATURES

Outstanding Freeze Start Capability
Industry-leading Durability

Functional Safety: ASIL C(Optional ISO26262:2018)
Software Architecture: AUTOSAR
Ingress Protection: IP67

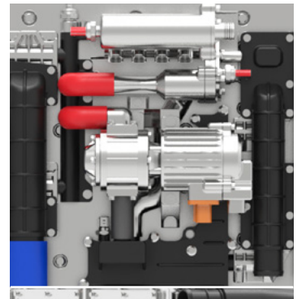
TECHNICAL SPECIFICATIONS

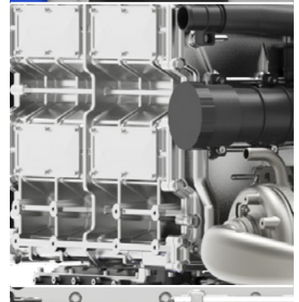
Rated Power Net (kW): 60
Freeze Start Temp. (°C): -30
Working Voltage (V): 450 - 750
Design Life (h): 15,000
Dimensions (mm): 995 x 818 x 593
Weight (kg): 210

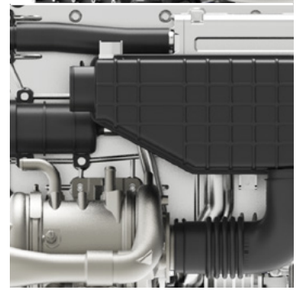


PRISMA XII+
Fuel Cell System

PRISMA XII+







DESCRIPTION

117kW Fuel Cell System
Heavy-Duty Applications

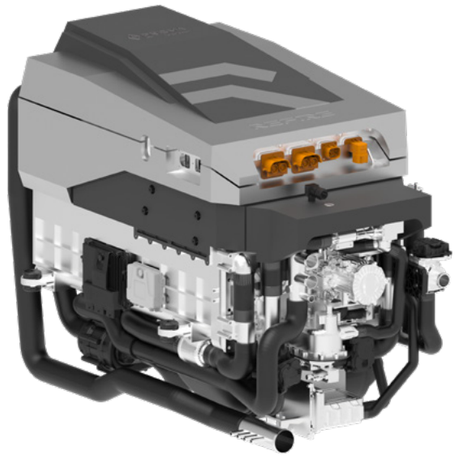
FEATURES

Outstanding Freeze Start Capability
Industry-leading Durability

Functional Safety: ASIL C(ISO26262:2018)
Software Architecture: AUTOSAR
Ingress Protection: IP67

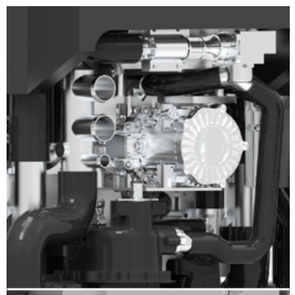
TECHNICAL SPECIFICATIONS

Rated Power Net (kW): 117
Freeze Start Temp. (°C): -30
Working Voltage (V): 450 - 750
Design Life (h): 25,000
Dimensions (mm): 892 x 767 x 826
Weight (kg): 278

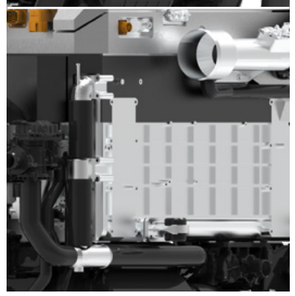


PRISMA XXII
Fuel Cell System

PRISMA XXII







DESCRIPTION

220kW Fuel Cell System
Heavy-Duty Applications

FEATURES

Outstanding Freeze Start Capability
Industry-leading Durability
Air Compressor With Expander, Exhaust Energy Recycling

Functional Safety: ASIL C(Optional ISO26262:2018)
Software Architecture: AUTOSAR 4.2
Ingress Protection: IP67
EMC: Level 3

TECHNICAL SPECIFICATIONS

Rated Power Net (kW): 220
Freeze Start Temp. (°C): -30
Maximum continuous operating water temperature(°C): 95
Rated Efficiency(%): 45
Peak Efficiency(%): 60
Weight(kg): 335 w. DC

SHOWCASING

ENDLESS POSSIBILITIES >>

Together with our farsighted, early adopter partners, REFIRE has always strived to take prototypes to production and to roll these out at scale in order to serve industry and the community as rapidly as possible. In 2018, beginning with a fleet of 500 7.5t trucks for freight forwarders in Shanghai, the city set the stage for other regions to witness what was possible with hydrogen fuel cell power. This also necessitated writing the framework for public usage of hydrogen as an energy carrier.

Since then, we have continued to work with more partners to deploy hydrogen fuel cell-powered vehicles and power machines.

The following is a small selection of the many projects completed by REFIRE. Please contact us for details of the 100+ fuel cell powertrain projects and prototypes commissioned by our customers. Or tell us about a zero-emission project for which you would like help finding solutions - we would love to help.



CLEAN LOGISTICS HYBATT BUS (BRANDENBURG) >>

On July 29th 2021, German bus operator UVG took its first delivery of a fuel cell bus, powered by a REFIRE fuel cell system. The bus was commissioned to operate routes between the town of Schwedt and the ecological tourist area known as the Lower Oder Valley National Park, in the state of Brandenburg.

Launched in response to the EU's Clean Vehicle Directive, the fuel cell bus project has been warmly welcomed by residents and officials from the area who praised its clean and quiet operation, zero emissions power, and impressive range of over 300km.





HQUBE 49T TRUCK >>

Equipped with REFIRE's PRISMA fuel cell system, the 49-tonne HQUBE heavy duty truck delivers a peak power of 126kW. The vehicle houses 12 hydrogen storage tanks pressurized to 70Mpa which can be refuelled in only 5-8 minutes.

Thanks in part to its streamlined, low drag coefficient cab design the HQUBE is capable of nearly 1000km on a single fill. This range ensures that it meets the commercial requirements of heavy duty, long haul, intercity and inter-provincial freight.







YUTONG 12M BUS >>

In the winter of 2019, the city of Zhengzhou, in the central plains of China, put 150 Yutong REFIRE fuel cell-powered buses into operation. Drivers of the 12-metre-long vehicles began a new chapter in zero-emissions mass transit during the harshest weather conditions of the year. No fewer than 37 systems were subjected to rigorous performance tests, component optimization, electromagnetic safety, and noise reduction protocols in order to meet Yutong's stringent engineering standards.

The fleet of environmentally friendly buses operate on twenty routes across the city. Drivers and passengers routinely comment on the pleasant user experience and the contribution to pollution reduction.



FAW 18T SPRINKLER >>

Sprinkler trucks across China are about to be revolutionized with non-polluting drivetrains. Sanitation vehicles typically cleanse city streets during the warmer months, but emissions from conventional diesel models are an unwanted and concentrated atmospheric burden, particularly because of the need for slow operation.

Manufactured by FAW, the 18T sprinkler has been reinvented by REFIRE to incorporate a zero emissions hydrogen-electric powertrain. When hydrogen fuel cells combine hydrogen and fresh air to produce on-demand electric power, the only by-product is pure water. Additionally, the low noise produced by the upgraded vehicles is a valuable benefit to urban residents.



DONGFENG 7.5T LOGISTIC VEHICLE >>

500+ Dongfeng logistics trucks deployed by REFIRE made up the world's largest commercial zero carbon hydrogen-powered fleet when they were launched in 2017. Since then, these vehicles have been helping over 20 Chinese e-commerce enterprises to implement green urban logistics across Shanghai and neighbouring cities such as Kunshan, Suzhou, Hangzhou, Changshu, and even as far afield as Yangshan Deepwater Port.

Today the vehicles continue to reliably serve the region, helping to clean the air, reduce noise, and conserve energy.



SANY 31T CEMENT MIXER >>

The urbanization of China has be extraordinarily rapid with ever-changing sky-lines and landscapes. The humble cement mixer is a vehicle right at the centre of these changes. It must navigate infrastructure and commercial development pro-ject sites, often in adverse conditions. Durability and reliability are prerequisites. Every component must interoperate flawlessly, from the chassis and suspension to the fuel cell powertrain.

REFIRE Application Engineering has redefined this workhorse of the construction industries with a zero carbon operational footprint. All the benefits of its hydro-gen-electric drive are brought to bear on the complex duty cycle, such as impres-sive torque, quiet operation, driver comfort and ease of use. This vehicle exemplifies REFIRE's ambition to green the most polluting sectors first.



FAW J6P >>

Since its debut in 2021, REFIRE's 49-tonne FAW heavy-duty tractor has been making waves in intercity logistics. Drivers and fleet operators benefit from the fast refueling time of just 15 minutes and the 300+ kilometre range even when fully loaded. This is made possible by the high energy density and power efficiency of the fuel cell system which is optimised for best performance under anticipated use case conditions.

Ease of operation and simple transfer of driver skills from Diesel to the FAW's fuel cell powertrain enable this truck to have immediate positive ecological impact over typical intercity routes.





4.5T LAST MILE CITY DELIVERY TRUCK >>

Jointly developed by REFIRE, the 4.5-tonne refrigerated city delivery truck has been designed for full integration of hydrogen storage tanks and the fuel cell-electric powertrain into the chassis without reduction in cargo volume. With a range of around 500km, it represents an improvement over pure BEV (Battery Electric Vehicle), and a viable clean alternative to Diesel-powered refrigerated trucks.

This vehicle will make its debut in 2022 Q4, providing a quiet and green solution to urban distribution of food and goods requiring refrigeration. By 2025 it is expected that 5,000 of these vehicles will be operating in several of the most densely populated regions of China. It will also be offered to customers in Europe and the Americas to help accelerate the transformation of logistics to clean hydrogen-electric power.

CONTACT US

THANK YOU! >>

We value your interest in REFIRE and in the hydrogen-electric energy revolution. Whether or not you are already involved, we would love to tell you more about our plans for decarbonisation of mobility and power, so the world can breathe easy.

Please email or call us to start the conversation.

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